National Junior College
SH2 Preliminary Examinations for General Certificate of Education Advanced Level
Higher 2

GEOGRAPHY **9173/02**
Paper 2

16 September 2025

3 hours

Additional Materials: Insert

READ THESE INSTRUCTIONS FIRST
An answer booklet will be provided with this question paper. You should follow the instructions on the front cover of the answer booklet. If you need additional answer paper, ask the invigilator for a continuation booklet.

Write your Centre number, index number and name on the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Section A
Answer **Question 1**.

Section B
Answer **Question 2.**

Section C
Answer **one** question.

The number of marks is given in brackets [ ] at the end of each question or part question.

Answer Questions 1 and 2 and
EITHER Question 3 **OR** Question 4.

Section A

Cluster 4: Fieldwork

1 A group of six students in Singapore carried out fieldwork to investigate how to engage youth in recycling efforts and encourage them to adopt environmentally sustainable practices through textile recycling and upcycling. The hypothesis for their investigation was:

'Youth aged between 15 and 35 years old residing in Serangoon are more likely to participate in textile recycling and upcycling if they have greater awareness of the environmental impact of the textiles industry and textile waste.'

To test their hypothesis, they decided to observe people who deposited their clothes into the textile recycling bins at three chosen sites in an urban neighbourhood, Serangoon, on a Friday during the June holidays. The students split themselves into three groups, and collected data at the three sites concurrently. The neighbourhood was chosen due to its proximity to their school.

At each site, the students sat nearby to record the total number of people who deposited their clothes and noted the number of youth after asking them if they were aged between 15 and 35.

By using convenience sampling, they administered a survey to 30 respondents aged between 15 and 35 at the three sites, with 10 respondents per site, to better understand their level of awareness of the environmental impact of the textiles industry and textile waste, as well as their level of participation in textile recycling and upcycling. They obtained consent and veified their ages before starting the survey. At the same time, an in-depth interview was conducted with six of the respondents to gain deeper insights. Based on their data collected, they concluded their hypothesis was valid.

Resource 1A is a map showing the locations of the three chosen sites in Serangoon marked as X, Y and Z. Resource 1B shows photographs of the three chosen sites. Resource 2 shows the data collected at each site. Resource 3 shows the questionnaire survey used by the students. Resource 4 shows selected results of

the questionnaire survey. Resource 5 shows selected excerpts of the in-depth interviews.

(a) Explain why this hypothesis is appropriate for the fieldwork. [4]

(b) With reference to Resources 1A and 1B, explain the strengths and weaknesses of the students' choice of sites and neighborhood. [6]

(c) Explain the strengths and limitations of the sampling method used by the students. [4]

(d) Using Resource 2, sketch a graph to present the data and justify your choice. [5]

(e) With reference to Resource 3, explain the strengths of the survey form used. [6]

(f) Suggest how the students could improve the accuracy and reliability of their data in Resource 4. [5]

(g) With reference to all the resources, evaluate the validity of their conclusion. [10]

Section B

Cluster 3: Sustainable Future and Climate Change

2 The Arcadis Sustainable Cities Index (SCI) 2024 provides an in-depth analysis of urban sustainability across 100 cities worldwide. It evaluates their performance across four core pillars: *Planet, People, Profit,* and the newly introduced *Progress* pillar, to assess how cities are advancing towards sustainable future.

Resource 6 shows the ranking of Top 3 Cities based on the SCI in 2024. Resource 7 shows a photograph of Kinshasa, the capital of Democratic Republic of Congo. Resource 8 is a fact file on Democratic Republic of Congo.

(a) Define sustainable urban development. [2]

(b) Describe how the indicator of Sustainable Cities Index can be used to measure sustainable urban development. [5]

(c) Compare the ranking of the cities shown in Resource 6. [6]

(d) Suggest reasons for the ranking described in **(c)**. [5]

(e) With reference to Resources 7 and 8, explain the possible challenges in progressing towards sustainable urban development in Kinshasa. [7]

(f) Explain how the vulnerability of different groups of people may result in varied impacts of climate change. [5]

Section C

Answer **EITHER** Question 3 **OR** Question 4.

3 ‘Creating cities that are liveable is more achievable than making them truly sustainable.’

Evaluate this statement. [20]

4 ‘The countries least responsible for climate change often suffer its harshest consequences and face challenges in implementing effective responses.’

Evaluate this statement. [20]

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